Multi-Scale Analysis of Terrestrial Vegetation Phenology Using Wavelet and Empirical Mode Decomposition Methods

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***Abstract*—This paper reports the results of a term project which investigated the application of time-frequency analysis methods to terrestrial vegetation phenology. The discrete wavelet transform (DWT), the continuous wavelet transform (CWT), and empirical mode decomposition (EMD) were used to analyze vegetation indices obtained from the Moderate Resolution Imaging Spectroradiometer (MODIS) over a 20-year period for three sites in Michigan. For each of the three analysis methods, phenology timing metrics were extracted from the time-frequency domain and compared against the existing MODIS Land Surface Dynamics product (MCD12Q2.061) as an independent validation reference. The DWT and CWT methods agreed on peak greenness timing within 2 days of each other on average across all sites, while EMD showed less consistency, and all methods exhibited site-dependent biases relative to MCD12Q2.061.**

***Index Terms*—Phenology, Remote Sensing, Vegetation Indices, Time-Frequency Analysis, Multi-resolution Analysis, Wavelet Transform, Empirical Mode Decomposition.**

I. Introduction

Phenology is the study of periodic events in ecological cycles and how intra- and inter-annual variations in climate drive these events. These ecological cycles include cyclical fluctuations of any biological system such as migratory patterns, colony development, or the emergence of vegetation. [1] As such, phenological signals are inherently non-stationary and multi-scale making them ideal subject matter for time-frequency analysis. This term project targeted the analysis of terrestrial vegetation phenology using the Enhanced Vegetation Index (EVI) derived from multispectral surface reflectance measurements. Classical methods for determining vegetation phenology metrics, such as number of vegetation cycles, the onset of greenness, and senescence, employ threshold-based and curve-fitting approaches on smoothed or reconstructed time series data rather than exploiting the time-frequency structure of these signals [2]. While the application of wavelet decomposition to phenological signals is well-studied [3] [4] [5] [6] [7] [8], direct extraction of phenological timing metrics from the time-frequency domain remains sparse in the literature. This project applied the discrete wavelet transform (DWT), continuous wavelet transform (CWT), and empirical mode decomposition (EMD) to capture annual signal components and extract phenological timing metrics directly from these components. These timing metrics were then compared against the MODIS Land Surface Dynamics product (MCD12Q2.061) [9] as an independent validation reference.

II. Materials

This work selected the EVI over the Normalized Difference Vegetation Index (NDVI) for analysis due to its ability to better correct for atmospheric conditions and canopy background noise as well as its higher sensitivity in areas with dense vegetation [10].

A. Site Selection

EVI time series data was obtained for years 2001 through 2020 for three sites in Michigan. A low latitude, mid latitude, and high latitude site were chosen to account for variability in both vegetation and duration and timing of the growing season (see Table I).

B. Data

EVI grid data was acquired from the MODIS Terra/Aqua Vegetation Indices (MOD13A1.061/MYD13A1.061) products via Microsoft Planetary Computer [11] [12] [13]. Combined, the MODIS Aqua and Terra satellites provide a temporal resolution of 8 days. For each site, a 3-by-3 grid of 500-meter pixels was extracted from the EVI grid data. First, pixels contaminated by clouds, ice/snow, and shadows were masked out. Next, the 3-by-3 grid was spatially averaged then snapped to an 8-day grid to ensure a constant sampling period. Finally, missing data values were filled using Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) interpolation. Fig. 1 plots the EVI time series data for each site resulting from the described data retrieval and conditioning pipeline.

TABLE I
Sites Summary

Site	Vegetation Type	Latitude	Longitude
A	Cropland	42.23405	-84.84961
B	Mixed Forest	44.59190	-84.42740
C	Evergreen Needleleaf Forest	46.30960	-86.59370

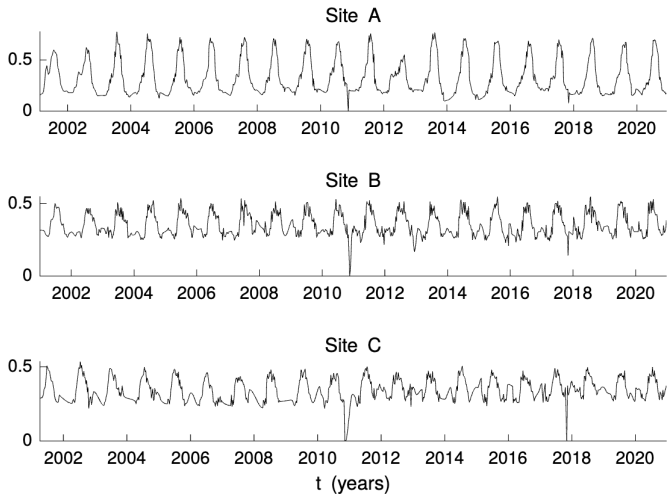

Fig. 1. EVI shown for (top) the low latitude site, (middle) the mid latitude site, and (bottom) the high latitude site.

III. Methodology

The first objective of this project is to demonstrate how the EVI time series data can be decomposed into inter-annual, annual, and sub-seasonal components using different methods. The second, and primary, objective of this project is to extract phenological timing metrics directly from the annual scale components obtained via these decompositions.

A. Analysis

Since the primary objective of this project is timing metric extraction, the wavelet families for the DWT and CWT analyses were selected according to conventions established in the environmental sciences literature in which the Meyer wavelet is the standard choice for DWT analysis [4] [5] and the Morlet wavelet is the standard choice for CWT analysis [6] [7]. An evaluation of the impact of wavelet family choice on analysis and timing metric extraction is deferred to future work.

1) Discrete Wavelet Transform: The DWT was used to perform multiresolution analysis (MRA) on the EVI time series data according to the methodology laid out by [4] and [5]. The MRA successively decomposes the EVI signal into approximation and detailed components at different scales. In order to extract the annual scale component, a relationship between these scales and period must be defined (see Table II). One can use the dominant, or center, frequency of the wavelet to define a periodic signal at scale j with period [14]

$$p_j = \frac{2^{N-j} T_s}{\nu_c}$$

where N is the maximum decomposition level, T_s is the sampling period, and ν_c is the center frequency of the wavelet. Here, the scale j is numbered according the subspace nesting convention

$$V_j \subset V_{j+1},$$

the maximum decomposition level of the three signals of length $L \approx 900$ is $N = 9$, and the center frequency of the Meyer wavelet is $\nu_c \approx 0.6634$. The scale most closely corresponding to the annual scale is $j = 4$ with a period of $p_4 = 386$ days.

TABLE II
Scale-Period Relationship

Scale (j)	**Decomposition Level**	**Period** (p)
0	9	6175
1	8	3087
2	7	1544
3	6	772
4	5	386
5	4	193
6	3	96
7	2	48
8	1	24

The first relevant output of this MRA is the total intra-annual variability

$$V(t) = \sum_{j=4}^{7} w_j(t)$$

where $w_j(t)$ is the time domain reconstruction of the detailed coefficients at scale j and $w_8(t)$ is discarded as high-frequency noise. A second relevant output is the inter-annual trend component $c_3(t)$ where $c_j(t)$ is the time domain reconstruction of the approximation coefficients at level j. Finally, the output that is key to the analysis of this project is the annual component $w_4(t)$ which will be used as input to the timing metric extraction portion of this work as done similarly in [4]. Fig. 2 plots these outputs for the three sites.

2) Continuous Wavelet Transform: The CWT is known to produce more data redundancy and be more computationally expensive than the DWT. However, it is preferred for analysis tasks requiring more precision in the time domain and is the standard choice for performing *wavelet coherence analysis*, a

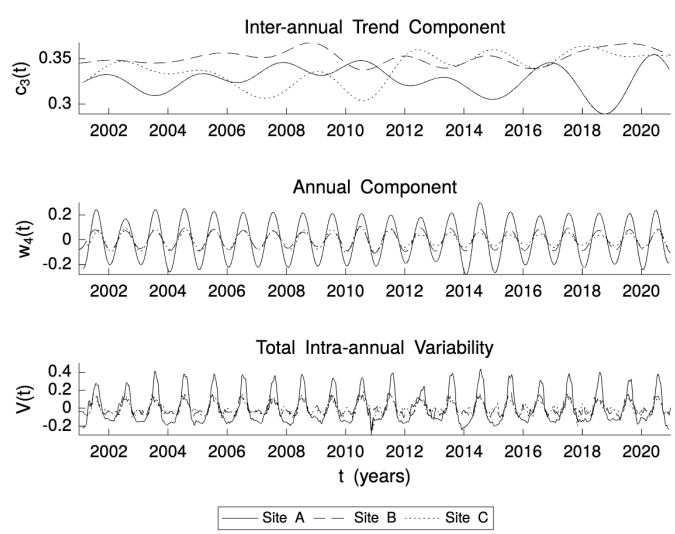

Fig. 2. The inter-annual trend components (top), annual components (middle), and total intra-annual variability components (bottom) shown for the three sites.

valuable technique for identifying environmental drivers across multiple time scales and an extension of this work of great professional interest [6] [7].

The Morlet wavelet produces a complex-valued transform $X(a,b) \in \mathbb{C}$ for scale a and position b, a key feature of this CWT analysis which enabled decomposition of the EVI time series into the periodic components $\mathrm{Re}\{X(a,b)\}$, $\mathrm{Im}\{X(a,b)\}$, and $\angle X(a,b)$ where $\mathrm{Re}\{X(a,b)\}$ produces positive peaks in phase with the peaks of the EVI signal. The CWT was computed using 32 voices per octave allowing for a near-perfect isolation of the annual scale component, an option not provided by the DWT. Two distinct annual scale components, $\mathrm{Re}\{\tilde{X}(a_0,b)\}$ and $\angle\tilde{X}(a_0,b)$, were extracted at the scale $a = a_0$ where the period at this scale is $p_{a_0} = 363$ days, the period most closely corresponding to the annual scale, and $\tilde{X}(a,b)$ is the region of the scalogram inside the *cone of influence*. These components (shown in Fig. 3) are used as inputs to the timing metric extraction.

3) Empirical Mode Decomposition: Unlike the DWT and CWT, EMD is a fully data-adaptive decomposition method that does not require a choice of fixed basis functions. EMD decomposes a signal into a finite set of *intrinsic mode functions* (IMF) via an iterative sifting process where each IMF represents a physically meaningful oscillatory component at a different instantaneous frequency [15]. A known limitation of EMD is mode mixing, a side effect of the sifting process in which a single IMF contains widely disparate scales or a single scale is split across multiple IMFs. Ensemble EMD (EEMD) was used to suppress mode mixing by averaging the IMFs obtained from N_e independent EMD trials where each trial was performed on the signal corrupted by white noise $\mathbf{w} \sim \mathcal{N}(0,\sigma_n^2)$ [16]. An ensemble size of $N_e = 200$ and noise standard deviation of $\sigma_n = 0.2$ was selected following the recommendations of [16] which showed that residual IMF noise decreases as $\sigma_n/\sqrt{N_e}$, reducing the noise contribution to approximately 1.4% of the signal standard deviation for the chosen parameters. Algorithm 1 summarizes the EEMD procedure. Next, the Hilbert transform was used to calculate the instantaneous period of each IMF, and finally, the annual scale component $c_a(t)$ used as input to the timing metric extraction was computed as

$$c_a(t) = \sum_{k:\bar{p}_k \in [180,540]} c_k(t)$$

where $\bar{p}_k$ is the average period of $c_k(t)$. For the three sites examined in this work, the number of obtained IMFs ranged from five to seven.

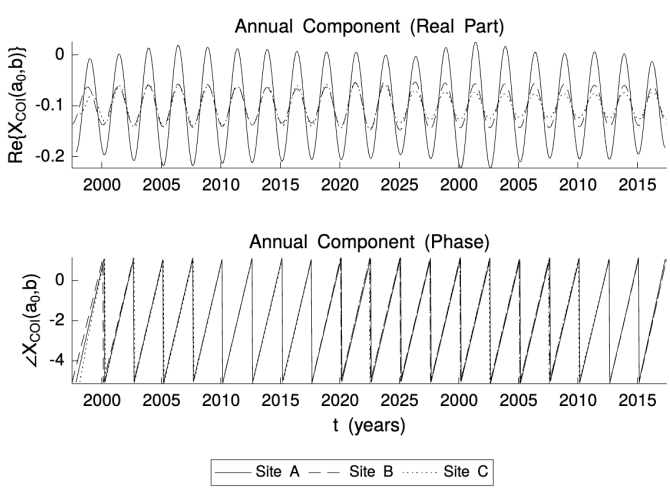

Fig. 3. The two annual scale components, (top) $\mathrm{Re}\{\tilde{X}(a_0,b)\}$ and (bottom) $\angle\tilde{X}(a_0,b)$, extracted from the CWT.

Algorithm 1 Ensemble Empirical Mode Decomposition

Require: $x(t)$, N_e, σ_n
Ensure: Ensemble-averaged IMFs $\{\bar{c}_k(t)\}$
Run EMD trial on $x(t)$ to obtain the number of IMFs K
Initialize accumulator $S_k(t) \leftarrow 0$ for $k = 1,\ldots,K$
for $i = 1$ **to** N_e **do**
  $w(t) \sim \mathcal{N}(0,\sigma_n^2)$
  $z(t) = x(t) + w(t)$
  Run EMD on $z(t)$ to obtain IMFs $\{c_k^{(i)}(t)\}_{k=1}^K$
  $S_k(t) \leftarrow S_k(t) + c_k^{(i)}(t)$ for $k = 1,\ldots,K$
end for
Compute $\bar{c}_k(t) = S_k(t)/N_e$ for $k = 1,\ldots,K$
return $\{\bar{c}_k(t)\}_{k=1}^K$

B. Timing Metric Extraction

For each of the three sites, eight phenological timing metrics (see Table III) were extracted directly from the four annual scale components established in Section III-A. All metrics were extracted within a phenological year window beginning on the 320th day of the year to avoid calendar year artifacts that would otherwise split the winter dormancy period across two windows. Thresholds were applied to the amplitude-normalized annual component on the ascending and descending limbs independently. The four extraction methods are summarized in Table IV.

IV. Evaluation

The MODIS Land Surface Dynamics product (MCD12Q2.061) was used as an independent validation reference [9]. This product estimates the same eight metrics tabulated in Table III by fitting piecewise logistic functions

TABLE III
Timing Metrics

Timing Metric	Definition
No. Cycles	Number of peaks with prominence $\geq 10\%$.
Greenup	20% threshold crossing of ascending limb.
Greenup Mid	50% threshold crossing of ascending limb.
Maturity	80% threshold crossing of ascending limb.
Peak	Maximum of annual component.
Senescence	80% threshold crossing of descending limb.
Greendown Mid	50% threshold crossing of descending limb.
Dormancy	Minimum of annual component.

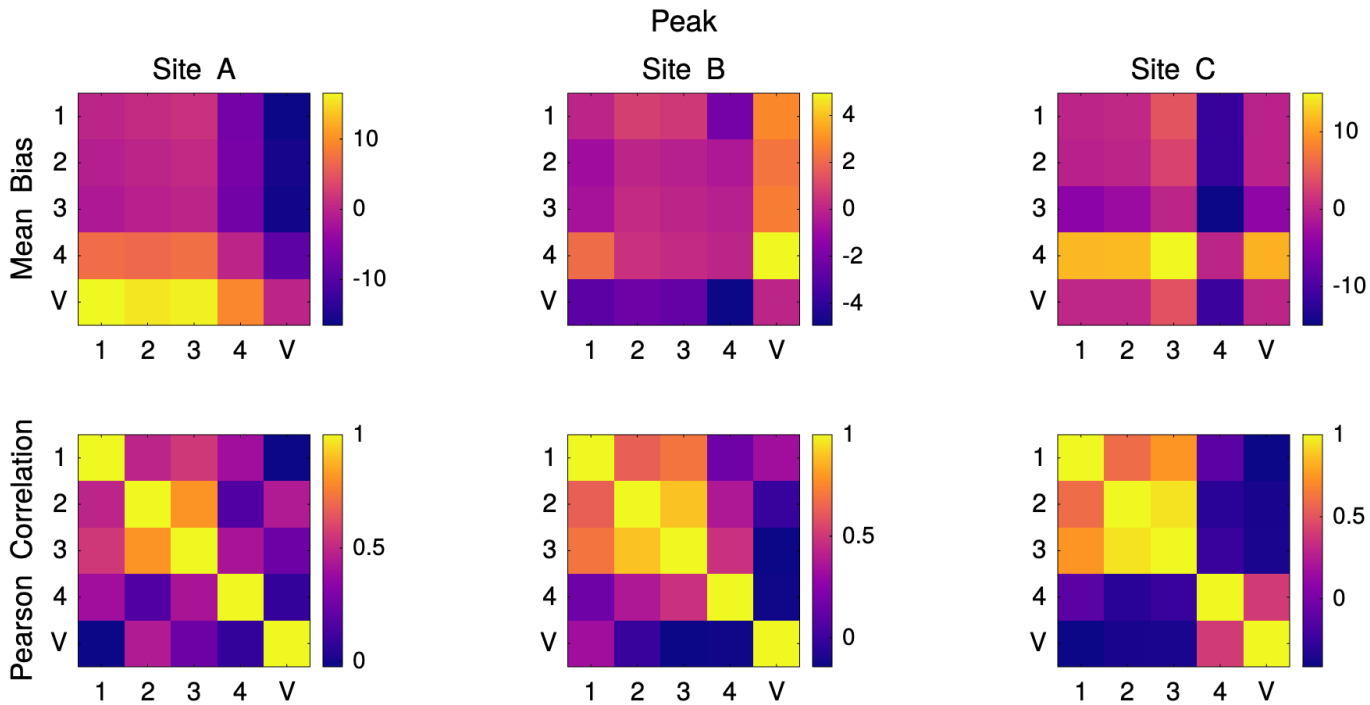

Fig. 4. The mean bias $\bar{B}$ (top) and Pearson correlation coefficient (bottom) between Methods 1-4 and the validation reference (indicated by "V") set shown for the three sites.

to MODIS EVI data. This work reports results for the "Peak" metric as it is uniquely defined as the maximum of the annual component making it the metric most amenable to cross-method comparison.

Fig. 4 plots the mean bias $\bar{B}$ and Pearson correlation coefficient ρ between Methods 1–4 and the validation reference for the "Peak" metric across the three sites. The DWT and CWT methods (Methods 1, 2, and 3) show the lowest bias with values between 0.25 and 2 days for Sites A and B and between 0.5 and 5 days for Site C. The small inter-method bias may be explained by differences in time-frequency localization between the Meyer and Morlet wavelets as consistent with the time-frequency uncertainty principle, though rigorous verification is deferred to future work. Method 4 does not consistently agree with Methods 1, 2, and 3 for any site.

The DWT and CWT methods report peak greenness approximately 2 weeks earlier than the validation data for Site A, approximately 2–3 days later for Site B, and within 1 day for Site C (with the exception of Method 3). Method 4 shows inconsistent agreement with the validation data, with bias magnitudes between 9 and 12 days.

TABLE IV
TIMING METRIC EXTRACTION METHODS

Method	Transform	Annual Component
1	DWT	$w_4(t)$
2	CWT	$\mathrm{Re}\{\tilde{X}(a_0, b)\}$
3	CWT	$\angle\tilde{X}(a_0, b)$
4	EMD	$c_a(t)$

V. CONCLUSION

Each of the DWT and CWT analysis methods successfully decomposed the MODIS EVI time series into physically interpretable sub-seasonal, annual, and inter-annual components, while empirical mode decomposition showed less consistency and reliability relative to the validation data. The project code and data pipeline are publicly available at https://github.com/doironty/ece866-term-project.

Future work includes the following methodological improvements and natural extensions to broader applications:

1) Theoretically relate and verify systematic offsets between methods to the time-frequency localization of the wavelet choices.
2) Diagnose the source of the inconsistent offsets between the four methods and the validation data.
3) Explore the choice of different wavelets for each of the DWT and CWT methods and compare against field data to determine which method is most physically meaningful.
4) Extend this framework to inform a multi-scale forecasting pipeline for both aquatic and terrestrial vegetation phenology.

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